MD SADIK RAHMAN

Full-Stack & Backend Software Engineer | CSE Undergraduate

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OBJECTIVE

Computer Science & Engineering student specializing in scalable backend architectures, multi-tenant enterprise platforms, and real-time AI systems. Experienced engineering production full-stack systems with Node.js, Express.js, Laravel 11, TypeScript, PostgreSQL (pgvector), and MongoDB. Proven track record building voice-activated commerce with Pipecat WebRTC and enterprise CRM workflows with 12-role RBAC. Seeking a Software Engineer / Backend Developer Internship to engineer high-throughput, secure, and maintainable systems.

TECHNICAL SKILLS

Backend & APIs:	Node.js, Express.js, Laravel 11, PHP 8.3, TypeScript, RESTful API Design, WebSockets, WebRTC, JWT Auth, Spatie RBAC, Rate Limiting
AI & Real-Time:	Pipecat AI, Google Gemini Live API, LangChain, Vector Embeddings (1536-dim), Semantic Search (pgvector), WebRTC Audio Pipeline
Databases & Cloud:	PostgreSQL 16 (pgvector, JSONB snapshots), MongoDB, Mongoose, Redis 7, Cloudinary Media Pipeline, Railway, Vercel
Frontend & UI:	React 18/19, Next.js, Vite, FilamentPHP v3 (Blade, Livewire 3, Alpine.js), Tailwind CSS, Three.js / R3F, Framer Motion
Tools & DevOps:	Docker & Docker Compose, Nginx, Git, GitHub, Postman, Playwright E2E, Linux, PM2, npm, Bun
Core & Languages:	C, C++, JavaScript (ES6+), TypeScript, PHP, Python, Data Structures & Algorithms (DSA), OOP

EDUCATION

BSc in Computer Science and Engineering **2025 – 2029 (Expected)**

Mymensingh Engineering College, Bangladesh

Higher Secondary Certificate (HSC) | GPA: 5.00 / 5.00 **2024**

Secondary School Certificate (SSC) | GPA: 5.00 / 5.00 **2022**

FEATURED PROJECTS

BazzarBD – AI Voice Commerce & Dual-Database E-Commerce Platform **2026**

Tech: React 18, Vite, TypeScript, Node.js, Express.js, PostgreSQL (pgvector), MongoDB, Pipecat AI, Gemini Live, WebRTC, Tailwind CSS

Live: **thelagaribd.com** | Code: **Sadik386/bazzarbd**

- Architected a dual-database e-commerce engine pairing MongoDB for catalog & transactional orders with PostgreSQL `pgvector` for 1536-dimensional semantic search and sub-100ms similarity retrieval.
- Integrated an interactive conversational AI Voice Salesman (Bangla & English) powered by Pipecat WebRTC and Google Gemini Live API with real-time screen and page-context awareness.
- Engineered voice-driven end-to-end checkout with atomic inventory reservation, automated location-based delivery charge calculation (inside/outside Dhaka), and Cash on Delivery / bKash workflows.
- Built an administrative merchant dashboard for inventory management, dynamic catalog CRUD, discount coupons, and automated courier fulfillment (Pathao & Steadfast).

Venture Visionary CRM – Enterprise Multi-Tenant ERP & Visa Operations Platform **2026**

Tech: Laravel 11, PHP 8.3, FilamentPHP v3, PostgreSQL 16, Redis 7, Docker Compose, Nginx, Tailwind CSS, Playwright

Client: **Venture Visionary Consultancy (Doha, Qatar)** | Architecture: **Enterprise Blueprint v2.0**

- Engineered an enterprise multi-tenant ERP platform supporting 3 dedicated portals (Staff Admin `/staff`, B2B Partner `/partner`, Client Portal `/portal`) governed by 12-role Spatie RBAC and strict tenant data isolation.
- Developed an omnichannel lead ingestion pipeline handling webhooks from Meta Ads, WhatsApp, and web inquiries with Round-Robin agent routing and deduplication across Qatar ID (QID), passport, and phone numbers.
- Built a 4-stage Quality Control (QC) Document Vault (`requested → uploaded → under_review → verified / rejected`) and a dynamic 193-country visa requirements engine with immutable JSONB snapshots.
- Implemented a financial ledger with itemized profit margin splits (Government/Embassy vs VAC vs Consultancy fees), wholesale B2B prepaid credit wallets, and Maker-Checker dual authorization.

PROBLEM SOLVING & COMPETITIVE PROGRAMMING

- **Codeforces (Handle: Sadik386):** Solved 70+ algorithmic problems focusing on implementation, number theory, and greedy algorithms.
- **Data Structures & Algorithms:** Strong proficiency in C++ STL (vectors, maps, sets, queues), recursion, and complexity analysis (O(n) time/space trade-offs).

LANGUAGES & AVAILABILITY

Bangla: Native **English:** Professional **Urdu:** Fluent **Hindi:** Conversational

Availability: Available for Internship and Junior Software Engineer roles (Remote / Hybrid / On-site). Flexible scheduling aligned with academic commitments.